Cyclistic User Behavior Analysis: From Casual Riders to Annual Members of 2025

Prepared by: Amin Suleiman

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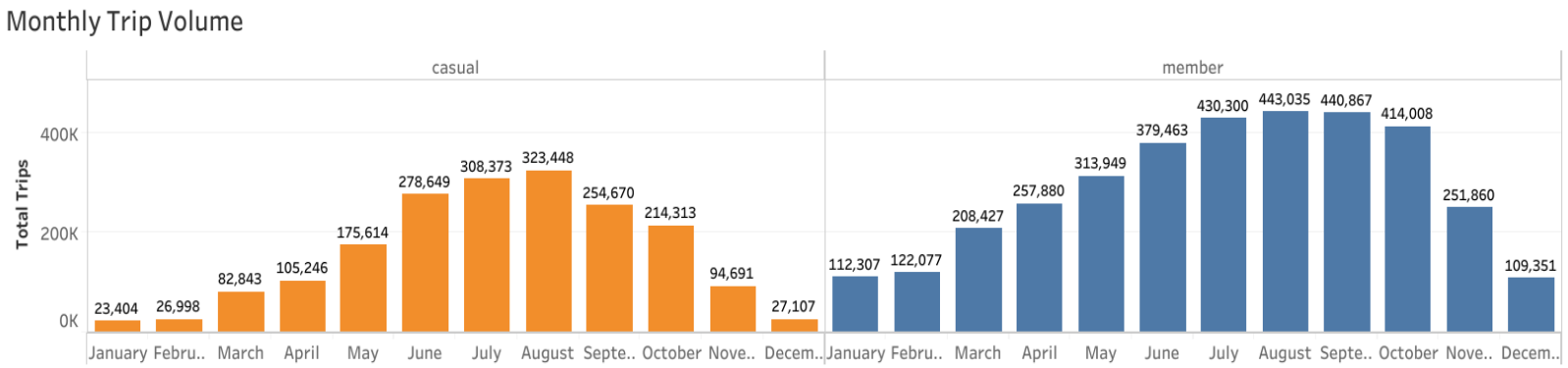
Business Task: Strategic Membership Growth

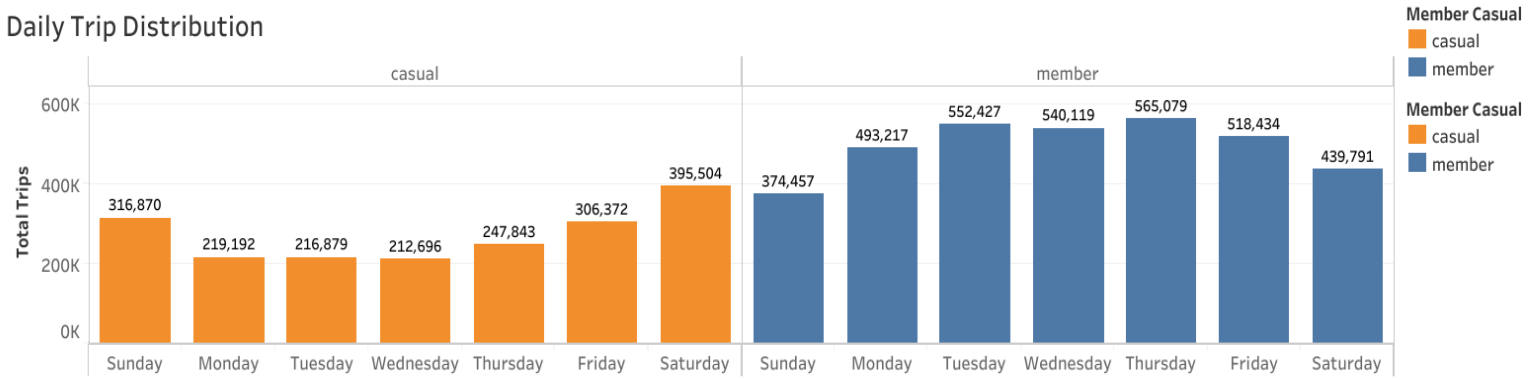
- Objective: Analyze behavioral differences between annual members and casual riders to design a data-driven conversion strategy.
- Core Problem: Annual members are more profitable; future growth depends on maximizing memberships.
- Key Goal: Transition from broad marketing to targeted strategies that convert casual riders into annual members.
- Analytical Focus: Identify trends in ride duration, timing, and frequency to identify conversion opportunities.
- Success Metric: Deliver data insights and visualizations to secure executive approval for marketing recommendations.

Data Preparation & Technical Workflow

- Data Source: Cyclistic's historical trip data (Public data provided by Motivate International Inc.).
- Processed 5.8M+ rows (12 months) via SQL for unified cleaning and merging.
- Data Cleaning: Removed invalid rides (<1m or >24h), validated data to ensure no duplicates,outliers,entry errors.
- Null Handling: Excluded 1M+ records from station rankings; retained for behavioral metrics.
- Feature Engineering: Generated Ride Length, Hour, Day, and Month dimensions.
- Visualization: Developed executive Tableau dashboards for trend analysis and insights.

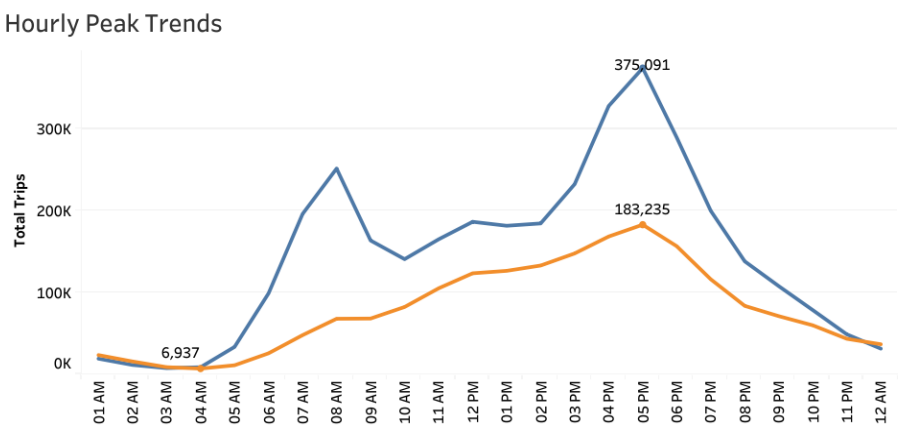
Key Performance Metrics: Casual vs. Member

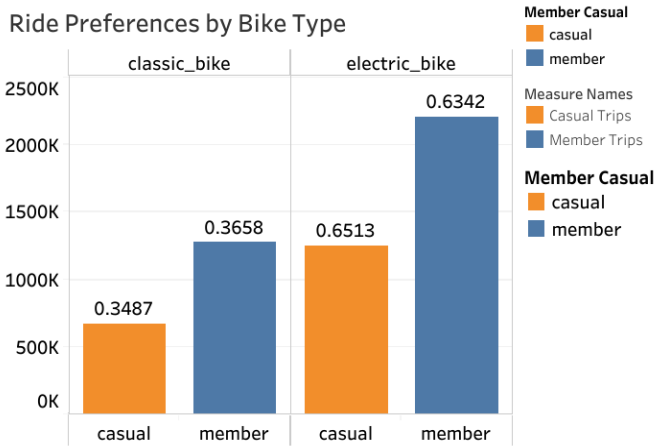


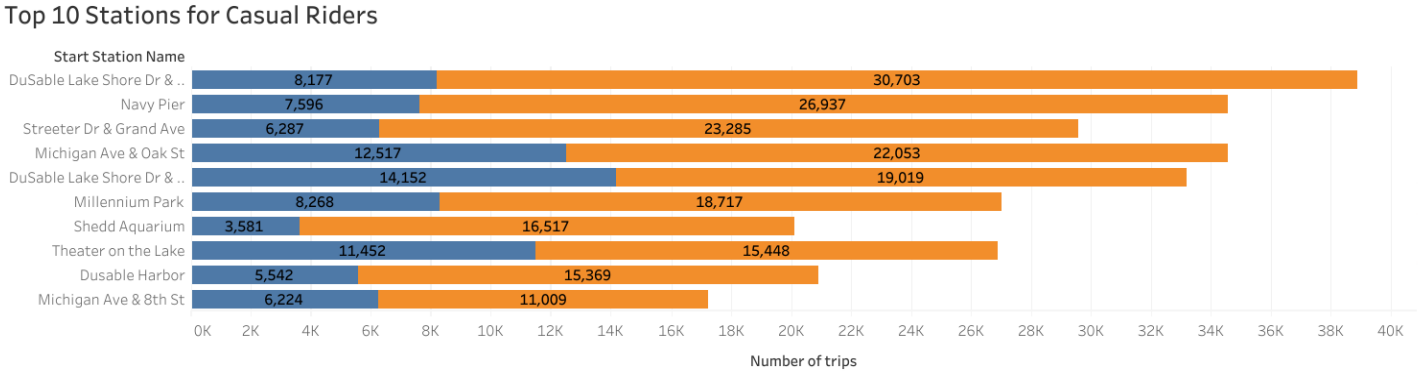


- **Seasonal Peak: Casual ridership peaks in August with over 323K trips, a 13x increase from January.**
- **Weekend Dominance: Casual trips reach ~395K on Saturdays, outperforming their weekday average by 64%.**
- **Commuter Core: Members maintain a steady ~500K+ trips throughout the work week (Monday-Friday).**
- **Growth Opportunity: 47% of casual trips occur during peak summer months(June,July,August), representing the primary conversion window.**

Behavioral Insights: Peak Hours & Station Traffic

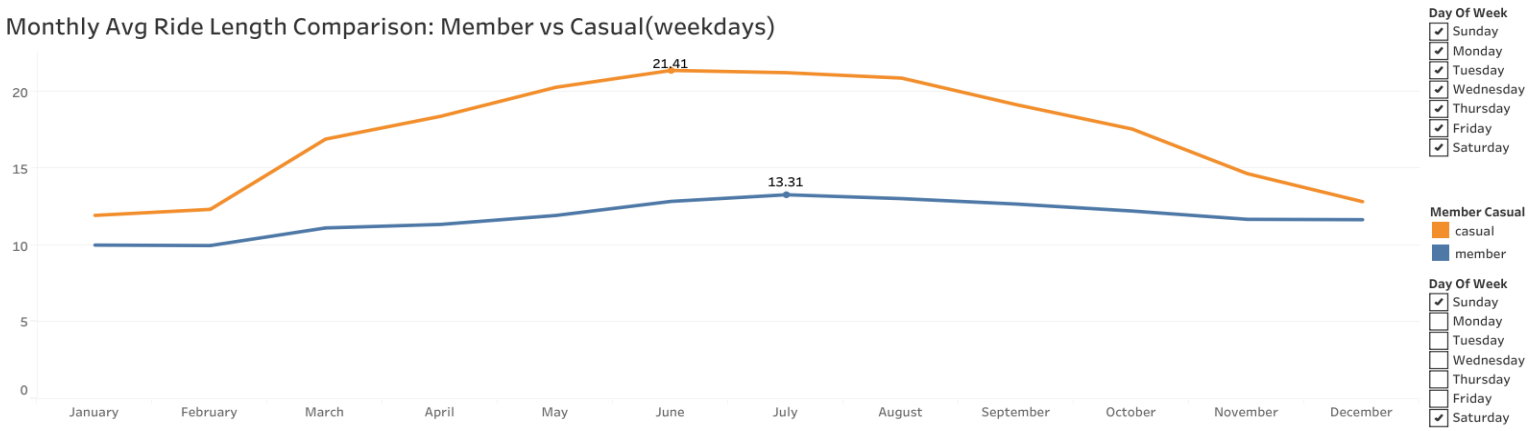


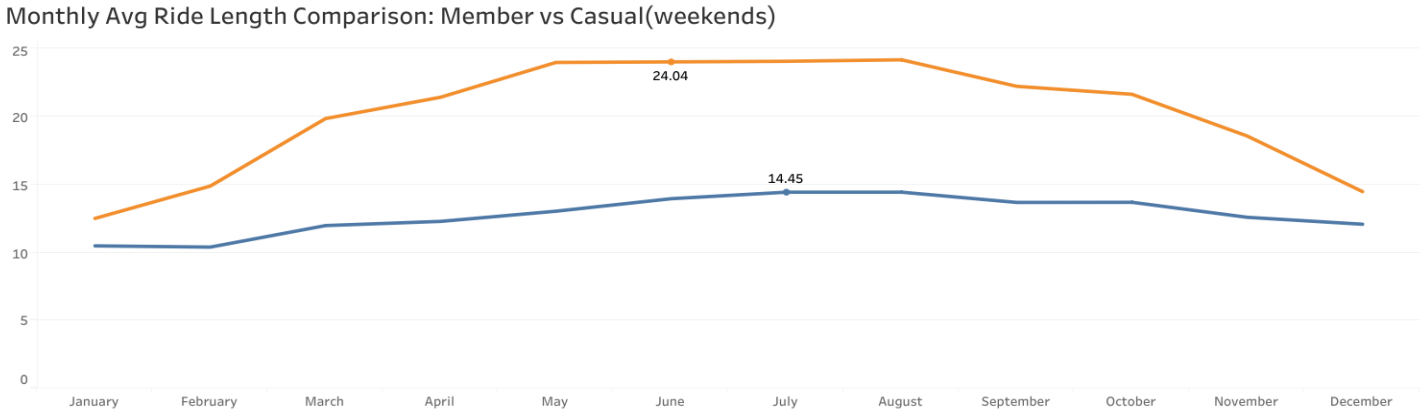




- **Commuter Peaks: Annual members show a massive spike at 5 PM (375K+ trips), confirming their use for daily work commutes.**
- **Leisure Patterns: Casual ridership builds steadily throughout the afternoon, peaking between 12 PM and 5 PM.**
- **High-Traffic Hubs: Dusable Lake Shore Dr & Monroe St is the top station for casual riders, highlighting the need for targeted weekend marketing at tourist locations.**
- **Casual riders prefer electric bikes.**

Trip Duration: Weekdays vs. Weekends





- **Weekday Efficiency: Annual members maintain a consistent 12-minute average, while casual riders average 21.4 minutes, even on working days.**
- **Weekend Leisure Peak: During August weekends, casual trips jump to a 24.04-minute average, nearly double the member duration.**
- **Target Opportunity: The significant gap in trip duration confirms that casual riders prioritize the experience and sightseeing, offering a prime conversion angle.**

Strategic Recommendations for Conversion

1) Implement Behavior-Based Conversion Triggers:

Target high-potential casual riders (3+ monthly rides or 90+ weekend minutes) with automated, personalized membership discounts (10–15%) via the Cyclistic app and email.

2) Execute Geo-Targeted Weekend Campaigns:

Deploy geo-fenced digital promotions at the Top 10 leisure stations during peak weekend hours (12 PM–5 PM), offering immediate membership credit for that day's ride.

3) Launch an Electric-Bike Membership Bundle:

Create a membership tier or bundle that reduces the per-ride electric surcharge, specifically targeting casual riders who use electric bikes 2+ times per month to position membership as a cost-efficient upgrade.

Strategic Implementation Note:

To maximize ROI, it is highly recommended to prioritize the execution of these strategies during the peak summer season (June through August), when casual rider engagement is at its highest.

Estimated Business Impact (Modeled Scenario)

- Total annual casual trips: 1.9M+ rides
- Estimated annual casual rider base: 380K–500K users (modeled range based on average ride frequency assumptions)
- Assumed annual membership price: $120

Revenue Projection Scenarios:

Conservative Case – 3% Conversion:

11,400–15,000 new members

Projected annual revenue increase: $1.3M–$1.8M

Target Case – 5% Conversion:

19,000–25,000 new members

Projected annual revenue increase: $2.3M–$3.0M

Strategic Note:

While revenue projections are based on full-year activity , conversion campaigns should prioritize peak summer months (June–August), where engagement is highest. This ensures annual strategic alignment with focused, high-ROI execution.

Executive Summary: Strategic Growth Path

- **Targeted Conversion:**

Casual riders show repeat behavior aligned with early membership patterns — a scalable, data-backed conversion opportunity.

- **Seasonal Activation:**

Activate precision digital campaigns during peak summer and weekend periods.

- **Product Differentiation:**

Leverage electric-bike preference to strengthen premium positioning.

- **Projected Financial Upside:**

Estimated revenue impact: $1.3M–$3M annually.

Thank You & Next Steps

- Final Thought: Data shows that casual riders are not just "temporary users"—they are leisure enthusiasts ready for a premium membership experience.

- Contact for Feedback:

Amin Suleiman

Data Analytics Project 2025

Questions?